AI & SOCIETY COURSE

Prof. Mor Naaman

Part 5

AI'S WORLDWIDE IMPACT

Unit 2: AI in Education

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Where We Are in the Course

Part 1 — AI Foundations
1. What is Generative AI: How it Works and What it Can Do
2. Bias and Limitations: Where GenAI Fails

Part 2 — AI and Individuals
3. AI and You: Cognition, Reliance, and Creativity
4. Persuasion by Design: How AI Influences You

Part 3 — AI, Communication, and Information
5. AI-Mediated Communication and Relationships
6. AI, Culture, and the Information Environment

Part 4 — Challenges for AI Implementation
7. Trust and Safety: AI Harms and Prevention
8. Data, Privacy, and IP: Who Owns What
9. The Challenging Economics of AI

Part 5 — AI's World-wide Impact
10. AI and the World: Education, Work, Power, Democracy
11. Externalities: Human and Environmental Costs of AI

Part 6 — The Way Forward with AI
12. The Future: Navigating AI in Your Community

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This unit: goals

- Understand the impact of generative AI on education and learning
- Think about the way forward: what students, teachers, and universities should do about it

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Two narratives

GenAI is a powerful educational tool

GenAI is destroying education

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The opportunity: a personal tutor for everyone

- Any level, any topic: helps student at their level, at their own pace, in their own language
- Always available, patient, and (nearly) free



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The opportunity: a personal tutor for everyone

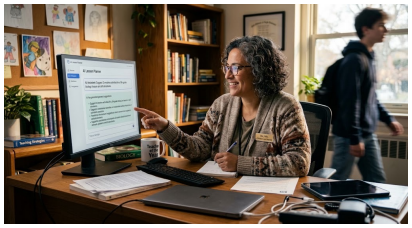
- Any level, any topic: helps student at their level, at their own pace, in their own language
- Always available, patient, and (nearly) free
- Example — language learning:
  - Unlimited conversation practice with instant correction, and no embarrassment in front of a class
  - The scarce resource used to be a patient fluent speaker with time for you.
- Example — learning to code:
  - Explains error messages, reviews your code, and produces worked examples on demand
  - But: the same tool will happily write the whole assignment for you

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The opportunity: a personal teacher's assistant

- For teachers: lesson plans, personalized/specialized materials, a first pass at feedback on materials, help in evaluations, …



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Let's look at the evidence: how is AI used (or misused) by students?

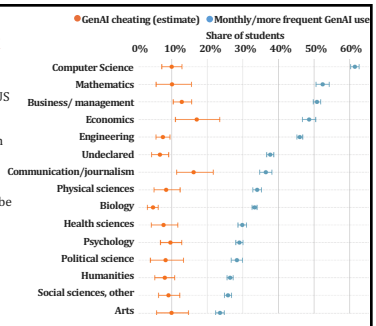
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AI use and misuse at scale

Survey of 95,513 students at 20 major US universities (in 2024!)

- **37%** use it monthly or more (62% in computer science, 24% in the arts)
- **9%** of GenAI users submitted AI-generated work knowing it may not be allowed (26% among daily users)
- Adoption is highest in Science/Tech; but cheating is highest outside it (economics 17%, journalism 16%)



Source: Chirikov et al. (2026), "Generative AI use and misuse call for assessment reform"

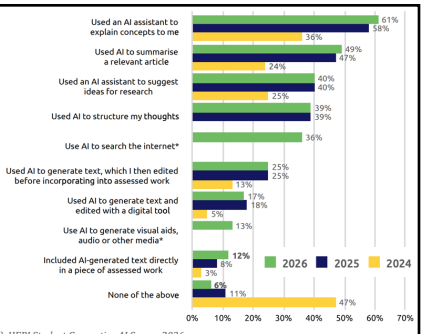
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AI use and misuse at scale

A survey of 1,054 undergraduates in the United Kingdom (in 2026)

"95% of students report using AI in at least one way and 94% say they use generative AI to help with [homework]."

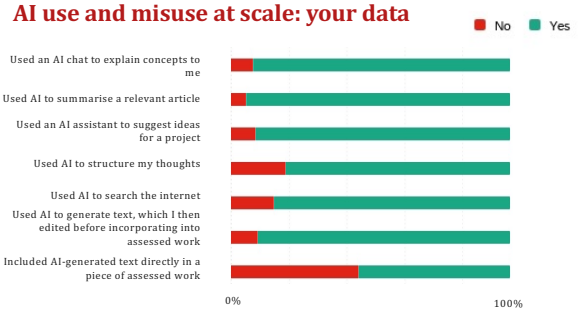


Source: Stephenson & Armstrong (2026), HEPI Student Generative AI Survey 2026

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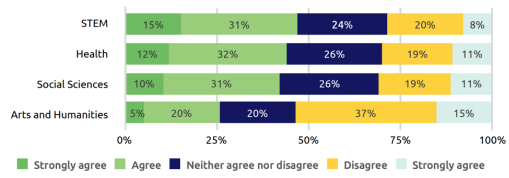
AI use and misuse at scale: your data



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"Content created by generative AI would get a good grade in my subject"

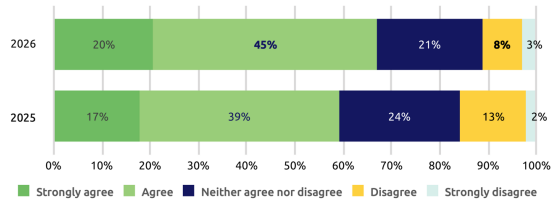


Source: Stephenson & Armstrong (2026), HEPI Student Generative AI Survey 2026

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"The way I am assessed by my institution has changed a lot in response to generative AI"



Source: Stephenson & Armstrong (2026), HEPI Student Generative AI Survey 2026

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Throwback: Unit 2-1, what AI does to learning

High-school math: better during practice with a ChatGPT tutor, worse on the exam without it

More Evidence from Science

High-school students learning math did better on practice problems with a ChatGPT tutor... but worse on the exam without it

Practice	Exam
▲ ChatGPT better than control	▼ ChatGPT worse than control

Also: students who used ChatGPT thought they performed better on the exam than those who didn't!

Slide credit: Prof. Andrew Perfors
Source: Bastani et al. (2025), "Generative AI without guardrails can harm learning: Evidence from high school mathematics"

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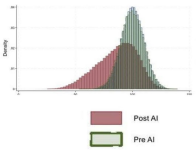
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Throwback: Unit 2-1, what AI does to learning

26,000 students in China. Homework scores up, exam scores down: the learning penalty

The Generative AI Learning Penalty



Study of 26,000 students in China measured casual impact of AI use on homework and exam scores

- ▲ Homework scores 18%
- ▼ Completion time 30%
- ▼ Monthly exam scores fell 20%
- ▼ high-stakes entrance-exam scores dropped 18–24%.

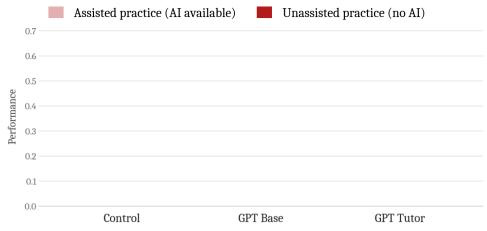
Source: Strömberg et al. (2026), "The Generative AI Learning Penalty: Evidence from Chinese Secondary Education"

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Results: performance with the AI, and without it

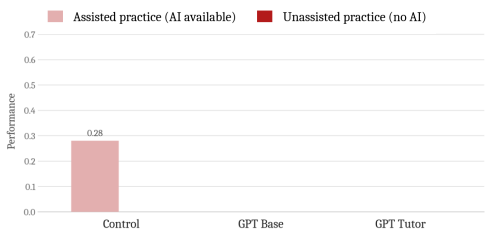


Source: Bastani et al. (2025), "Generative AI without guardrails can harm learning," PNAS

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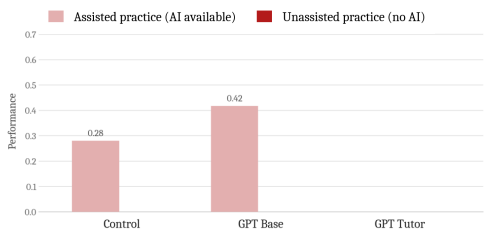


Source: Bastani et al. (2025), "Generative AI without guardrails can harm learning," PNAS

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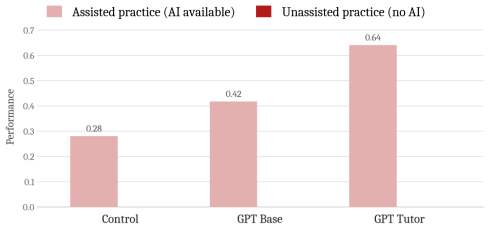


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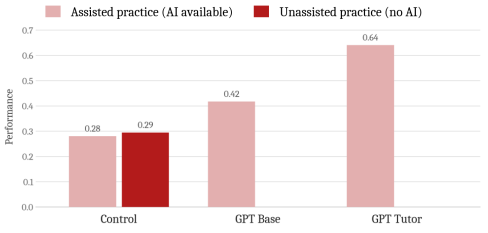


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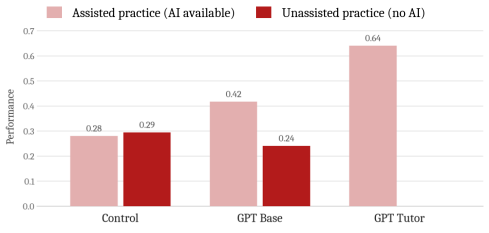


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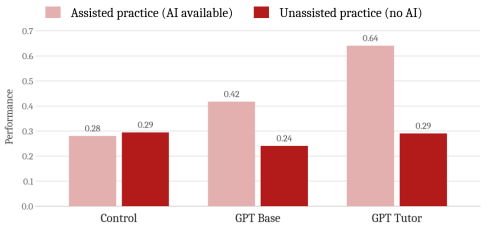


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Summary so far:

- Everyone is using AI, but adoption is likely uneven between study areas
- Many students admit to misusing AI (knowingly submitting AI work)
- AI is helping with grades but not helping with learning: not a good scaffolding!
  - But: carefully designed safeguards and tutors can help learning.
- Assessment has been changing!

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How should we approach genAI in education?

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Cornell's answer: lean in, lean out

- Cornell's framing: the decision is for each professor and course, not the university. Courses can mix different approaches!

Lean in
AI is part of the work. Students are assessed on the combined output, and possibly on how well they use AI.

Lean out
AI is not allowed, and the assessment is redesigned so it cannot be done with AI (for example, in the classroom).

- Course/assignments should say what is expected:

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Cornell's answer: lean in, lean out – examples



GenAI may be used [illegible] as specified. No other tools or [illegible] course concepts. [illegible] guidance.

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My city's answer: lean out (until high school) [good!]

"Under the rules, about 600,000 students in the city's schools — about two-thirds of all students in New York City public schools — will not be allowed to use A.I. until they enter high school, and even then, it will be permitted only in specific cases."


The New York Times

New York City Bans A.I. in Elementary and Middle Schools

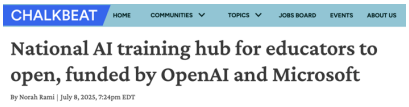
A new policy announced by the Department of Education bars younger students from using artificial intelligence, but stops short of a moratorium.

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Challenge: AI company's push

- One motivation: get the students early, so they get used to your tool and stay with it.
- Microsoft, OpenAI and Anthropic give $23M to provide free AI training for 400,000 teachers (July 2025)
- AI companies release tools for teachers (ChatGPT for Teachers, ChatGPT Edu, Claude for Teachers, and Gemini and NotebookLM)
- …


CHALKBEAT HOME COMMUNITIES TOPICS JOBS BOARD EVENTS ABOUT US

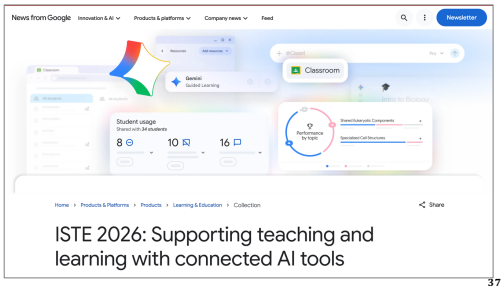
National AI training hub for educators to open, funded by OpenAI and Microsoft

By Noah Rami | July 8, 2025, 7:24pm EDT

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Challenge: AI company's push



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Answer 2: teach both, test both

- Teach students to work with AI in one part of the course: prompting, checking, verifying, knowing when the output is wrong
- Teach them to flourish without it in the other part of the course: the thinking that has to happen in their own head
- Test both, separately:
  - AI-allowed work, judged on delivery, judgement, verification and taste
  - AI-free work, judged on independent capability (in class, oral, or practical)
- Assessment-reform literature recommendation: no bans, no detection, but redesign

Credit: Daniel Suskind; Also see: Chirikov, Smirnov & Kizilcec (2026)

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But don't forget: it's all about motivation, for students and teaches!

- Are students are motivated by grade or by learning?
- Remember MOOCs?
- Teachers motivated by student learning need to redesign courses, assessments.
- No AI policy will work if there is a motivation mismatch.

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This unit: summary

- Two narratives: a personal tutor for everyone, or the end of learning
- Use is widespread and very uneven across disciplines.
- AI is not a great scaffolding by default, but can be used appropriately to enhance learning!
- The way forward is changing teaching and assessment to support "lean in" and "lean out" modes – maybe a mix!

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